

# [SOURCE OF TRUTH] -- Mechanical Dimensions and Specs

[LEV-72](#) · Project: Full Scale Integration · Milestone: Mechanical Integration  
Status: In Progress · Priority: Urgent · Assignee: Aditya Matam

## Dimensions and Integration Notes

*For mechanical teams — this is how much real-estate I need per yoke*

### Net BOUNDING BOX (worked out): 7" x 5.5" x 15"

- Yoke: **5" x 3.5" x 9"** (Image 1)
  - Mass: **14 kg**
- Sensor (Image 2): **4" away length-wise**
- Shielding box (that we place on top of the yokes during transport): **+2" on each side**
- Yoke + Shielding => 7" x 5.5" x 11" + Sensor Clearance => **7" x 5.5" x 15"**

### Forcing Details (informs ideal pod mass):

- There are 2 holes on either face of the yoke (Image 1). L-brackets interface the yoke to an **aluminum plate/beam** on the top of the pod (FRAME DEPENDENCY)
- **Forcing Approximation** from the script [here](#) and methodology described in [LEV-35](#)

Pod mass vs. equilibrium gap (6" yoke, 4 identical yokes, i=0, from `reluctance_model.py`):

| Equilibrium gap | Pod mass (kg) | Pod mass (lb) |
|-----------------|---------------|---------------|
| 5 mm            | 315.3         | 695.2         |
| 6 mm            | 275.7         | 607.8         |
| 7 mm            | 241.1         | 531.6         |
| 8 mm            | 211.0         | 465.1         |
| 9 mm            | 184.5         | 406.8         |
| 10 mm           | 161.3         | 355.6         |

6-8mm has been the "advised" equilibrium gap, giving us an ideal pod mass of **211-276 kg**.

*Note: These forcing figures assume 6" yoke, contradicting the bounding box figure I provided above. I would like mechanical teams to allot real estate for the 9" yoke in case we have to revert later.*

## Image 1 (Yoke)

*(see Linear ticket — signed image URL, not embedded here)*

## Image 2 (Sensor worst-case dimension margin)

*(see Linear ticket — signed image URL, not embedded here)*

## Integration Notes

Hi Aditya, I'm working on structures and dynamics and wanted to ask a few questions about the levitation system so we can account for its requirements in our design.

- What is the threshold for the pod weight that the levitation system can support? Is there an ideal weight we should aim for?
  - **Ideal mass would be 210-280 kg.** Counterintuitive, but the heavier side of that range is actually better for us.
- Are there any center of gravity requirements we should consider?
  - Equal weight distribution helps us **significantly**. If we can't achieve that because of electronics and such, perhaps we could consider putting some counterweights on the structure/chassis after-the-fact? Center of gravity is preferably low down, but I don't think it matters too much.
- Would having the wheel yokes on the outside of the pod cause any issues for your design?
  - Do you have a picture of what this looks like? I don't have a good visualization of the frame/track/wheels interaction.
- What spacing do the magnets and yokes need to function properly?
  - The bounding box I've set accounts for the spacing we need
- Are there any mounting points or clearances we should plan around?
  - Yes, see the pictures. If you need more (or actual CAD) let me know. My Solidworks is kinda broken, but I can get it to you.